# Circles & Coordinate Geometry

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: The equation of a circle**

The circle below has the radius shown. Find its area and circumference, in terms of $\pi$.

Area $= \pi(6)^2 = 36\pi$. Circumference $= 2\pi(6) = 12\pi$.

**2** **[Foundational]** • **Target: Circle theorems**

A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.

Inscribed angle $= \frac{1}{2}(100^\circ) = 50^\circ$.

**3** **[Foundational]** • **Target: Arc length and sector area**

The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of $\pi$.

$72^\circ = \frac{2\pi}{5}$ radians. Sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}(100)\left(\frac{2\pi}{5}\right) = 20\pi$.

**4** **[Foundational]** • **Target: The equation of a circle**

Write the equation of a circle with center $(-4, 7)$ and radius $9$.

$(x + 4)^2 + (y - 7)^2 = 81$.

**5** **[Foundational]** • **Target: The equation of a circle**

A circle has diameter $26$ and is centered at the origin. Write its equation, and state its radius.

Radius $= \frac{26}{2} = 13$. Equation: $x^2 + y^2 = 169$.

**6** **[Foundational]** • **Target: Completing the square**

Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.

$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16 = 36 \Rightarrow (x - 3)^2 + (y + 4)^2 = 36$. Center $(3, -4)$, radius $6$.

**7** **[Foundational]** • **Target: Completing the square**

Find the center and radius of $x^2 + y^2 + 10x - 4y + 13 = 0$.

$(x^2 + 10x + 25) + (y^2 - 4y + 4) = -13 + 25 + 4 = 16 \Rightarrow (x + 5)^2 + (y - 2)^2 = 16$. Center $(-5, 2)$, radius $4$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Distance, midpoint, and slope

Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.

$d = \sqrt{(4-(-3))^2 + (-7-5)^2} = \sqrt{49+144} = \sqrt{193} \approx 13.89$.

**9** [Intermediate] • Target: Distance, midpoint, and slope

Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.

Midpoint $= \left(\frac{2-8}{2}, \frac{-6+10}{2}\right) = (-3, 2)$. Slope $= \frac{10-(-6)}{-8-2} = \frac{16}{-10} = -\frac{8}{5}$.

**10** [Intermediate] • Target: Distance, midpoint, and slope

Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.

Slope $= \frac{-5-7}{6-(-2)} = \frac{-12}{8} = -\frac{3}{2}$. Since the slope is negative, the line falls from left to right.

**11** [Intermediate] • Target: Coordinate proofs

Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.

Slope of $AB = \frac{0-0}{4-0} = 0$ (horizontal). Slope of $BC$ is undefined (vertical, since $x = 4$ for both points). A horizontal segment and a vertical segment are perpendicular, so $\angle B = 90^\circ$, confirming a right triangle.

**12** [Intermediate] • Target: Coordinate proofs

Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.

Slope $AB = \frac{3-2}{4-1} = \frac{1}{3}$. Slope $DC = \frac{6-5}{3-0} = \frac{1}{3}$, so $AB \parallel DC$. Slope $AD = \frac{5-2}{0-1} = -3$. Slope $BC = \frac{6-3}{3-4} = -3$, so $AD \parallel BC$. Both pairs of opposite sides are parallel, so $ABCD$ is a parallelogram.

**13** [Intermediate] • Target: Tangent lines to circles

A circle has center $O(2, 3)$ and radius $5$. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from $P$ to the circle, to the nearest hundredth.

$OP = \sqrt{(2-2)^2 + (12-3)^2} = \sqrt{81} = 9$. Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{81-25} = \sqrt{56} = 2\sqrt{14} \approx 7.48$.

**14** [Intermediate] • Target: Tangent lines to circles

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.

The radius to $(3, 4)$ has slope $\frac{4}{3}$, so the tangent (perpendicular to the radius) has slope $-\frac{3}{4}$. Line: $y - 4 = -\frac{3}{4}(x - 3)$, which simplifies to $3x + 4y = 25$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Tangent lines to circles** • **Cross-Lesson: 7.01 Solid Geometry & Similarity**

A circle has center $(0, 0)$ and radius $10$. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.

Distance from center $= \sqrt{6^2 + 9^2} = \sqrt{117} \approx 10.82$, which is greater than the radius $10$. The point lies outside the circle.

**16** **[Challenging]** • **Target: Circle theorems** • **Cross-Lesson: 8.01 Statistics & Probability**

A central angle of $84^\circ$ subtends an arc. Find the measure of the inscribed angle subtending the same arc.

Inscribed angle $= \frac{1}{2}(84^\circ) = 42^\circ$.

**17** **[Challenging]** • **Target: Arc length and sector area** • **Cross-Lesson: 1.01 Advanced Algebra**

A circle has radius $12$ cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.

$s = r\theta = 12\left(\frac{5\pi}{6}\right) = 10\pi$ cm.

**18** **[Challenging]** • **Target: Arc length and sector area** • **Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic**

A sector of a circle with radius $9$ has area $27\pi$. Find the central angle, in radians.

$27\pi = \frac{1}{2}(81)\theta = 40.5\theta \Rightarrow \theta = \frac{27\pi}{40.5} = \frac{2\pi}{3}$.

**19** **[Challenging]** • **Target: Conics: ellipses and hyperbolas** • **Cross-Lesson: 3.01 Sequences, Series & Complex Numbers**

Identify the type of conic and state its center for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$.

An ellipse, centered at $(2, -1)$, with $a^2 = 16$ ($a = 4$, horizontal major axis) and $b^2 = 9$ ($b = 3$).

**20** **[Challenging]** • **Target: Conics: ellipses and hyperbolas** • **Cross-Lesson: 4.01 Lines, Triangles & Polygons**

Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$.

A hyperbola, centered at $(-3, 0)$, with $a = 2$, $b = 3$. Asymptote slopes: $\pm\frac{b}{a} = \pm\frac{3}{2}$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: The equation of a circle

Write the equation of a circle with center $(9, 12)$ and radius $12$.

**a.** $(x-9)^2+(y-12)^2=12$

**b.** $(x+9)^2+(y+12)^2=144$

**c.** $(x-9)^2+(y+12)^2=144$

**d.** $(x-9)^2+(y-12)^2=144$ ✓

$(x-9)^2+(y-12)^2=144$.

**22** [Foundational] • Target: Completing the square

Write the equation of a circle with center $(8, -7)$ and radius $7$.

**a.** $(x-8)^2+(y-7)^2=49$

**b.** $(x+8)^2+(y-7)^2=49$

**c.** $(x-8)^2+(y+7)^2=7$

**d.** $(x-8)^2+(y+7)^2=49$ ✓

$(x-8)^2+(y+7)^2=49$.

**23** [Foundational] • Target: Distance, midpoint, and slope

Write the equation of a circle with center $(11, -9)$ and radius $5$.

**a.** $(x+11)^2+(y-9)^2=25$

**b.** $(x-11)^2+(y+9)^2=25$ ✓

**c.** $(x-11)^2+(y+9)^2=5$

**d.** $(x-11)^2+(y-9)^2=25$

$(x-11)^2+(y+9)^2=25$.

**24** [Foundational] • Target: Coordinate proofs

Write the equation of a circle with center $(0, 10)$ and radius $10$.

**a.** $(x)^2+(y-10)^2=102$

**b.** $(x)^2+(y-10)^2=100$ ✓

**c.** $(x)^2+(y-10)^2=10$

**d.** $(x)^2+(y+10)^2=100$

$(x)^2+(y-10)^2=100$.

**25** [Foundational] • Target: Tangent lines to circles

Write the equation of a circle with center $(2, 5)$ and radius $12$.

**a.** $(x-2)^2+(y-5)^2=12$

**b.** $(x-2)^2+(y+5)^2=144$

**c.** $(x+2)^2+(y+5)^2=144$

**d.** $(x-2)^2+(y-5)^2=144$ ✓

$(x-2)^2+(y-5)^2=144$.

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**26** [Foundational] • Target: Circle theorems

Write the equation of a circle with center $(-1, 9)$ and radius $11$.

**a.** $(x+1)^2 + (y-9)^2 = 121$ ✓

**b.** $(x-1)^2 + (y+9)^2 = 121$

**c.** $(x+1)^2 + (y+9)^2 = 121$

**d.** $(x+1)^2 + (y-9)^2 = 11$

$(x+1)^2 + (y-9)^2 = 121$.

**27** [Foundational] • Target: Arc length and sector area

Write the equation of a circle with center $(11, -5)$ and radius $13$.

**a.** $(x+11)^2 + (y-5)^2 = 169$

**b.** $(x-11)^2 + (y+5)^2 = 13$

**c.** $(x-11)^2 + (y-5)^2 = 169$

**d.** $(x-11)^2 + (y+5)^2 = 169$ ✓

$(x-11)^2 + (y+5)^2 = 169$.

**28** [Foundational] • Target: Conics: ellipses and hyperbolas

Find the center and radius of $x^2 + y^2 + 2x - 2y - 62 = 0$.

**a.** $(-1, -1), 8$

**b.** $(-1, 1), 8$ ✓

**c.** $(1, 1), 8$

**d.** $(-1, 1), 10$

$(x^2 + 2x + 1) + (y^2 - 2y + 1) = 62 + 1 + 1 \Rightarrow (x+1)^2 + (y-1)^2 = 64$. Center $(-1, 1)$, radius $8$.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: A well-designed word problem: a Ferris wheel

Find the center and radius of $x^2 + y^2 - 6x - 2y - 90 = 0$.

**a.** $(3, 1), 12$

**b.** $(-3, 1), 10$

**c.** $(3, 1), 10$ ✓

**d.** $(3, -1), 10$

$(x^2 - 6x + 9) + (y^2 - 2y + 1) = 90 + 9 + 1 \Rightarrow (x-3)^2 + (y-1)^2 = 100$. Center $(3, 1)$, radius $10$.

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**30** [Intermediate] • Target: General Circle Equations & Completing the Square

Find the center and radius of $x^2 + y^2 - 8x - 18y + 81 = 0$.

**a.** $(-4, 9), 4$

**b.** $(4, -9), 4$

**c.** $(4, 9), 6$

**d.** $(4, 9), 4$ ✓

$(x^2 - 8x + 16) + (y^2 - 18y + 81) = -81 + 16 + 81 \Rightarrow (x - 4)^2 + (y - 9)^2 = 16$. Center $(4, 9)$, radius $4$.

**31** [Intermediate] • Target: Tangent Lines to Circles

Find the center and radius of $x^2 + y^2 - 10x + 8y + 37 = 0$.

**a.** $(5, 4), 2$

**b.** $(-5, -4), 2$

**c.** $(5, -4), 2$ ✓

**d.** $(5, -4), 4$

$(x^2 - 10x + 25) + (y^2 + 8y + 16) = -37 + 25 + 16 \Rightarrow (x - 5)^2 + (y + 4)^2 = 4$. Center $(5, -4)$, radius $2$.

**32** [Intermediate] • Target: Loci & Conic Definitions

Find the center and radius of $x^2 + y^2 + 4x + 14y - 91 = 0$.

**a.** $(-2, -7), 12$ ✓

**b.** $(2, -7), 12$

**c.** $(-2, -7), 14$

**d.** $(-2, 7), 12$

$(x^2 + 4x + 4) + (y^2 + 14y + 49) = 91 + 4 + 49 \Rightarrow (x + 2)^2 + (y + 7)^2 = 144$. Center $(-2, -7)$, radius $12$.

**33** [Intermediate] • Target: Coordinate Proofs & Polygons

Find the center and radius of $x^2 + y^2 + 12x - 18y + 17 = 0$.

**a.** $(-6, 9), 12$

**b.** $(6, 9), 10$

**c.** $(-6, 9), 10$ ✓

**d.** $(-6, -9), 10$

$(x^2 + 12x + 36) + (y^2 - 18y + 81) = -17 + 36 + 81 \Rightarrow (x + 6)^2 + (y - 9)^2 = 100$. Center $(-6, 9)$, radius $10$.

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**34** [Intermediate] • Target: The equation of a circle

Find the center and radius of $x^2 + y^2 - 2x + 4y - 116 = 0$.

**a.** $(1, -2), 13$

**b.** $(-1, -2), 11$

**c.** $(1, 2), 11$

**d.** $(1, -2), 11$ ✓

$(x^2 - 2x + 1) + (y^2 + 4y + 4) = 116 + 1 + 4 \Rightarrow (x-1)^2 + (y+2)^2 = 121$. Center $(1, -2)$, radius $11$.

**35** [Intermediate] • Target: Completing the square

A circle has center $O(-6, 7)$ and radius $5$. Point $P(-6, 20)$ lies outside the circle. Find the length of the tangent from $P$ to the circle.

**a.** 13

**b.** 14

**c.** 12 ✓

**d.** 10

$OP = \sqrt{((-6)-(-6))^2 + (20-7)^2} = \sqrt{169} = 13$. Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

**36** [Intermediate] • Target: Distance, midpoint, and slope

A circle has center $O(7, 5)$ and radius $5$. Point $P(7, 18)$ lies outside the circle. Find the length of the tangent from $P$ to the circle.

**a.** 13

**b.** 12 ✓

**c.** 10

**d.** 14

$OP = \sqrt{(7-7)^2 + (18-5)^2} = \sqrt{169} = 13$. Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

**37** [Intermediate] • Target: Coordinate proofs

A circle has radius $13$. Find the area of the sector with central angle $40^\circ$, in terms of $\pi$.

**a.** $\frac{338}{9}\pi$

**b.** $\frac{178}{9}\pi$

**c.** $\frac{160}{9}\pi$

**d.** $\frac{169}{9}\pi$ ✓

Sector area $= \frac{40}{360} \times \pi(13)^2 = \frac{40}{360} \times 169\pi = \frac{169}{9}\pi$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A circle has radius $12$. Find the area of the sector with central angle $150^\circ$, in terms of $\pi$.

**a.** $60\pi$ ✓

**b.** $61\pi$

**c.** $120\pi$

**d.** $59\pi$

Sector area $= \frac{150}{360} \times \pi(12)^2 = \frac{150}{360} \times 144\pi = 60\pi$.

**39** [Challenging] • Target: Circle theorems • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the distance between $(0, -7)$ and $(-1, 1)$, to the nearest hundredth.

**a.** $9.06$

**b.** $7.06$

**c.** $8.06$ ✓

**d.** $10.06$

$d = \sqrt{((-1) - 0)^2 + (1 - (-7))^2} = \sqrt{1 + 64} = \sqrt{65} \approx 8.06$.

**40** [Challenging] • Target: Arc length and sector area • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Find the distance between $(5, 9)$ and $(-2, 1)$, to the nearest hundredth.

**a.** $9.63$

**b.** $12.63$

**c.** $10.63$ ✓

**d.** $11.63$

$d = \sqrt{((-2) - 5)^2 + (1 - 9)^2} = \sqrt{49 + 64} = \sqrt{113} \approx 10.63$.

**41** [Challenging] • Target: General Circle Equations & Completing the Square • Cross-Lesson: 5.01 Trigonometry

A circle has the equation $x^2 + y^2 + 4x - 10y + 13 = 0$. What are its center and radius?

**a.** **Center** $(-2, 5)$**, radius** $4$ ✓

**b.** Center $(2, -5)$, radius $4$

**c.** Center $(-2, 5)$, radius $16$

**d.** Center $(2, -5)$, radius $16$

Complete the square: $(x + 2)^2 + (y - 5)^2 = -13 + 4 + 25 = 16$. Center is $(-2, 5)$ and radius is $\sqrt{16} = 4$.

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**42** [Challenging] • Target: General Circle Equations & Completing the Square • Cross-Lesson: 7.01 Solid Geometry & Similarity

For what value of $k$ does $x^2 + y^2 - 6x + 8y + k = 0$ represent a single point (degenerate circle)?

**a.** $25$ ✓

**b.** $16$

**c.** $9$

**d.** $0$

Completing square: $(x-3)^2 + (y+4)^2 = 9 + 16 - k = 25 - k$. For a single point, radius $= 0 \Rightarrow 25 - k = 0 \Rightarrow k = 25$.

**43** [Challenging] • Target: Tangent Lines to Circles • Cross-Lesson: 8.01 Statistics & Probability

At what points does the line $y = x + 2$ intersect the circle $x^2 + y^2 = 10$?

**a.** $(1, 3)$ **and** $(-3, -1)$ ✓

**b.** $(3, 1)$ and $(-1, -3)$

**c.** $(1, 3)$ only

**d.** No intersection

Substitute: $x^2 + (x+2)^2 = 10 \Rightarrow 2x^2 + 4x - 6 = 0 \Rightarrow x^2 + 2x - 3 = 0 \Rightarrow (x+3)(x-1) = 0$. Points are $(1, 3)$ and $(-3, -1)$.

**44** [Challenging] • Target: Loci & Conic Definitions • Cross-Lesson: 1.01 Advanced Algebra

What is the locus of all points whose distances to $(2, 0)$ and $(-2, 0)$ have a constant sum of $6$?

**a.** **Ellipse** $\frac{x^2}{9} + \frac{y^2}{5} = 1$ ✓

**b.** Hyperbola $\frac{x^2}{9} - \frac{y^2}{5} = 1$

**c.** Circle $x^2 + y^2 = 9$

**d.** Parabola $y^2 = 4x$

Constant sum $2a = 6 \Rightarrow a = 3$. Distance between foci $2c = 4 \Rightarrow c = 2$. $b^2 = a^2 - c^2 = 9 - 4 = 5$. Equation is ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$.

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**45** [Challenging] • Target: Coordinate Proofs & Polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

What type of quadrilateral has vertices at $A(0,0), B(4,0), C(5,3),$ and $D(1,3)$?

**a. Parallelogram ✓**

**b.** Rhombus

**c.** Rectangle

**d.** Trapezoid (not parallelogram)

Slope of $AB$ and $CD$ is $0$ (parallel, length $4$). Slope of $AD$ and $BC$ is $\frac{3}{1} = 3$ (parallel). Length $AD = \sqrt{1^2 + 3^2} = \sqrt{10} \neq 4$. Thus it is a parallelogram.

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